

4.13.16

EE25BTECH11011 - Navya

Question:

The lines $p(p^2 + 1)x - y + q = 0$ and $(p^2 + 1)^2x + (p^2 + 1)y + 2q = 0$ are perpendicular to a common line for

- | | |
|----------------------------|------------------------------|
| 1) exactly one value of p | 3) more than two values of p |
| 2) exactly two values of p | 4) no value of p |

Solution:

Given lines are,

$$p(p^2 + 1)x - y + q = 0 \quad (1)$$

$$(p^2 + 1)^2x + (p^2 + 1)y + 2q = 0 \quad (2)$$

The normal vectors of the lines are

$$\mathbf{n}_1 = \begin{pmatrix} p(p^2 + 1) \\ -1 \end{pmatrix}, \quad \mathbf{n}_2 = \begin{pmatrix} (p^2 + 1)^2 \\ p^2 + 1 \end{pmatrix} \quad (3)$$

If both lines are perpendicular to a common line with normal vector $\mathbf{m}$, then

$$\mathbf{n}_1^T \mathbf{m} = 0, \quad (4)$$

$$\mathbf{n}_2^T \mathbf{m} = 0. \quad (5)$$

Hence, the two equations can be written together as

$$\begin{pmatrix} p(p^2 + 1) & -1 \\ (p^2 + 1)^2 & p^2 + 1 \end{pmatrix} \mathbf{m} = \mathbf{0}. \quad (6)$$

For a non-trivial $\mathbf{m}$ to exist, the two rows must be linearly dependent.

$$\exists k \in \mathbb{R} \text{ such that } \begin{pmatrix} p(p^2 + 1)^2 & p^2 + 1 \end{pmatrix} = k \begin{pmatrix} p(p^2 + 1) & -1 \end{pmatrix}. \quad (7)$$

Equating components, we get

$$(p^2 + 1)^2 = k p(p^2 + 1), \quad (8)$$

$$p^2 + 1 = -k \quad (9)$$

From (9), we have

$$k = -(p^2 + 1) \quad (10)$$

Substituting into (8)

$$(p^2 + 1)^2 = -p(p^2 + 1)(p^2 + 1). \quad (11)$$

If $p^2 + 1 \neq 0$ dividing both sides by $(p^2 + 1)^2$ gives

$$1 = -p \implies p = -1. \quad (12)$$

If $p^2 + 1 = 0$, then $p = \pm i$, which are non-real.

Hence, both lines are perpendicular to a common line for exactly one real value of p .

$$\boxed{p = -1} \quad (13)$$

